

industry, they are opting for high-end insulated-gate bipolar transistor (IGBT) UPSes, which offer low harmonics and high power factor. They are also keen on reducing the carbon footprint and operating expenses by a target percentage set by themselves."

Earlier, in rural sectors, people were tolerant towards long power cuts. However, there has been a change in the behaviour pattern of people as they are now opting for backup solutions, notes Sood. She says, "People are also demanding efficient systems and the trend is shifting to solar-powered systems with maximum power point tracking (MPPT)."

Rao feels, customers have still not seen the actual benefit of solar inverters. He says, "It has a prolonged payback period of more than 10 years. Maybe three or five years down the line, we can expect solar inverters to trend but at this stage solar inverters are not in major demand."

On the other hand, Sivarajan notes, "When you are investing on a 500VA or 800VA solar inverter, the payback period might be a little longer but a solar panel lasts for a minimum of 25 years. Even if you are getting the return on investment after five years, it is still worth it." He adds, "Consumers are already reaping the benefits of solar inverters."

According to Rao, three- and four-level inverters are quite the trend in the market today. He says, "These inverters achieve very high switching speeds, low harmonic distortion and high reliability with a compact footprint." Maintenance is a key parameter that consumers consider today while choosing a secure power solution. He adds, "Advanced battery-management solutions and fire-proof batteries are some other trends in the UPS and inverter industry."

Three-level inverters are inverters with three levels of switching. Sivarajan says, "These provide slightly greater efficiency and nothing more than that." He adds, "The end customers, that is industrial or corporate users, do not exactly understand three- or four-level inverters. They look for a

The increasing penetration of IT/ITeS sector in tier II cities has boosted the UPS and inverter market

reliable power system that gives them backup whenever required." At the end of the day, the technology that the customer chooses has to serve the purpose of reliability.

Battery indicators in power backup systems is another trend picking up in the market to make the consumer aware of the battery life status. Sood says, "This is possible with inverters having battery management systems in place. Such solutions are already available in the market with tier I suppliers." She adds, "All inverter manufacturers are also looking to reduce the size of inverters by switching to high-frequency inverters. Such inverters would require smaller-size transformers, and will also help in reducing the overall cost other than improved efficiency."

Talking about high-frequency inverters, Sivarajan claims, "We have launched a new high-frequency inverter that has an efficiency of more than 90 per cent. If you consider a regular transformer based inverter, it provides a maximum efficiency of 80 to 82 per cent." He adds, "Technology is playing a heavy role in minimising the losses in inverters for better conservation and battery life."

Government policies

Recently, the solar energy sector was in for good news on the first day of the new year, as the Haryana government has issued a directive to have rooftop solar power system installed in every building of the plot size of 418.06sqm (500-square yards) or more. Systems are to be installed by September 2015, informs Sood. She says, "The order will be applicable to private bungalows, group housing societies, builder apartments, malls, offices, commercial complexes, schools, hospitals (any building, new or old,

that meets the plot-size criteria)." She adds, "Adding to the incentives for developers, the government will offer a 30 per cent subsidy on installation costs on first-come-first-served basis."

Also, with several state governments in India promoting the usage of solar power by announcing several policy changes, the hospitality and retail industry has shown early signs of adoption. Mukherjee says, "A number of vendors are working on solutions based on solar power. It will be interesting to observe the rest of the industrial community in India responding to solar-powered UPSes and inverters."

In some states, if a UPS that gives very high power factor to the source is deployed by manufacturing plants, IT organisations and other industrial facilities, the state incentivises their electrical consumption, informs Rao. He says, "There are also regulations coming from electricity boards to maintain the harmonics within certain limits. For a clean electricity environment, it is necessary for UPS manufacturers to adopt technologies such as IGBT. This benefits customers by not inducing high amount of harmonics to the source." He adds, "Our recommendation to the government would be to restrict measurable parameters, such as harmonics, to five per cent on the input side and power factor to 0.95."

Adding to the recommendations, Sivarajan says, "I would like the government policies to be 'Make in India' friendly so that Indian companies into manufacturing power backup hardware systems can benefit from better taxation system, infrastructure and road transportation system. There should be a simplified taxation system for a business to grow in India." "The initiatives towards building infrastructure to promote hardware industry from the new government looks good but it is too early to rate these," he adds.

Pricing trends

Prices of commodity items like copper, aluminium and steel are only increasing, so there are always challenges to keep the product competitive, feel